Blockchain Technology - Study Guide (MCQ)

Lecture 8 & 9: Public Blockchain & Bitcoin

Prepared for Study Purposes

Questions

1. **What is a core characteristic of a Public Blockchain?**
   - A) Restricted access to authorized users only.
   - B) Controlled by a central government authority.
   - C) Open network accessible to anyone without restrictions.
   - D) Requires permission from a central node to join.
2. **Which consensus algorithm was first proposed by Satoshi Nakamoto for Bitcoin?**
   - A) Proof of Stake (PoS)
   - B) Proof of Work (PoW)
   - C) Delegated Proof of Stake (DPoS)
   - D) Proof of Authority (PoA)
3. **What does the "51% Rule" in public blockchain security imply?**
   - A) Security depends on at least 51% of nodes being honest.
   - B) 51% of transactions must be free of charge.
   - C) Only 51% of the network can be active at any time.
   - D) A block is valid only if 51 nodes approve it.

4. **Which of the following is considered a disadvantage of Proof of Work (PoW)?**
    - A) High transparency.
    - B) Massive energy consumption.
    - C) Enhanced security.
    - D) Decentralization.
5. **What is the total fixed supply of Bitcoin?**
    - A) 10 million
    - B) 21 million
    - C) 50 million
    - D) Unlimited supply
6. **In Bitcoin, what is the purpose of a Private Key?**
    - A) To identify the user's public address.
    - B) To verify a transaction's signature.
    - C) To digitally sign transactions and prove ownership.
    - D) To calculate the mining difficulty.
7. **Which encoding method is used for Bitcoin addresses to make them human-readable?**
    - A) Base64
    - B) ASCII
    - C) Base58Check
    - D) Hexadecimal
8. **What does UTXO stand for in the context of Bitcoin?**
    - A) Universal Transaction X-change Output
    - B) Unspent Transaction Output
    - C) Unified Transfer eXternal Order
    - D) User Transaction Ownership
9. **What is the "Coinbase Transaction"?**

- A) The last transaction in a block.
- B) A transaction between two exchange platforms.
- C) The first transaction in each block, rewarding the miner.
- D) A transaction that has been rejected by the network.

10. **How many decimal places can a Bitcoin be divided into?**

    - A) 2 decimal places
    - B) 6 decimal places
    - C) 8 decimal places (Satoshis)
    - D) 18 decimal places

11. **Which field in the transaction data structure specifies the earliest time a transaction becomes valid?**

    - A) Version number
    - B) Lock time
    - C) Input counter
    - D) Script length

12. **What is the primary purpose of Transaction Fees?**

    - A) To pay for government taxes.
    - B) To incentivize miners and prevent network abuse.
    - C) To increase the total supply of Bitcoin.
    - D) To pay for the electricity of the sender.

13. **How is the Transaction Fee calculated?**

    - A) Fixed rate per transaction.
    - B) Sum(Outputs) - Sum(Inputs)
    - C) Sum(Inputs) - Sum(Outputs)
    - D) Based on the price of Bitcoin.

14. **What is the “Mempool”?**

    - A) A permanent storage for all blocks.

- B) A temporary storage for unconfirmed transactions.
- C) A pool where miners share their rewards.
- D) The database of all private keys.

15. **What is the main benefit of Segregated Witness (SegWit)?**

- A) It increases the total supply of Bitcoin.
- B) It separates digital signatures to reduce transaction size.
- C) It makes mining difficulty constant.
- D) It removes the need for private keys.

16. **How often is the Bitcoin mining difficulty adjusted?**

- A) Every block (10 minutes).
- B) Every 2016 blocks (approximately 2 weeks).
- C) Once a year.
- D) It never changes.

17. **What is the “Target” in Bitcoin mining?**

- A) The number of miners in the network.
- B) A specific value that the block hash must be lower than.
- C) The total amount of Bitcoin to be mined.
- D) The time it takes to mine one block.

18. **What happens if blocks are mined too quickly (less than 10 minutes)?**

- A) The difficulty decreases.
- B) The difficulty increases.
- C) The reward is doubled.
- D) The network shuts down.

19. **What is a “Mining Pool”?**

- A) A place where miners store their hardware.
- B) A group of miners who share their computational power and rewards.

- C) A list of all transactions waiting to be mined.
- D) A central server that controls all mining.

20. **Which algorithm is used twice during the Bitcoin address generation process?**
    - A) RIPEMD-160
    - B) ECDSA
    - C) SHA-256
    - D) Base58

Answer Key

Question	Answer
1	C
2	B
3	A
4	B
5	B
6	C
7	C
8	B
9	C
10	C
11	B
12	B
13	C
14	B
15	B
16	B
17	B
18	B
19	B
20	C